BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, SWAT

INTERMEDIATE (12TH CLASS) EXAMINATION, 2021

PHYSICS

Past Paper 2021

Retyped faithfully from the original board paper — same questions, no changes.

SECTION-A

Time Allowed: 20 Minutes | **Maximum Marks:** 18

Note: 1) Attempting all MCQs is compulsory. This paper along with the OMR sheet must be returned to the superintendent after due time. 2) Fill the circle A B C D, which one is correct with blue or black ball point in separate OMR Sheet like ● 3) If more than one circle in the OMR sheet is filled then no credit will be given to such answer.

Q1. Energy of electron in hydrogen 1st quantized orbit.

(A) 12.6ev
(B) 13.6ev
(C) 13.6kev
(D) None of these

Q2. The Magnetic flux will be maximum for an angle of.

(A) 0°
(B) 60°
(C) 90°
(D) 180°

Q3. The RMS value of current in an AC circuit is 10A, the peak value of current is.

(A) 14.1A
(B) 7.07A
(C) 0.707A
(D) 10A

Q4. The temperature at which a material becomes superconductor is.

(A) Curie temperature
(B) Critical temperature
(C) Neutral temperature
(D) Transition temperature

Q5. The De-Broglie wave length of an electron moving with a speed of 10^{6}m/s is.

(A) $7.3x10^{-10}$m
(B) $3.7x10^{-10}$m
(C) $7.3x10^{-9}$m
(D) $3.7x10^{-9}$m

Q6. The process of conversion of AC into DC is known as.

(A) Inversion
(B) Amplification
(C) Rectification
(D) Biasing

Q7. The torque in the coil can be increased by increasing.

(A) No. of turns
(B) Current and magnetic field
(C) Area of coil
(D) All of the above

Q8. The unit of magnetic flux is.

(A) Tesla
(B) Weber
(C) Electron volt
(D) Both A & B

Q9. Transistor is basically.

(A) Power amplifier
(B) Voltage Amplifier
(C) Energy amplifier
(D) None of these

Q10. Electric potential of earth is taken zero because earth is a good.

(A) Conductor
(B) Insulator
(C) Semi conductor
(D) None of these

Q11. When a wire is stretched and its radius becomes 1/2, then its resistance will be.

(A) 16 R
(B) 4 R
(C) 2 R
(D) 0

Q12. Ampere's law is applicable to.

(A) Circular path
(B) Rectangular path
(C) To any closed path
(D) None of these

Q13. Kirchhoff's voltage law is based on law of conservation of.

(A) Energy
(B) Charge
(C) Mass
(D) None of these

Q14. Nature of force b/w two current carrying wire in same directions.

(A) Attraction
(B) Repulsion
(C) Both Attraction and repulsion
(D) None of these

Q15. The device in which induced emf is statically induced emf is.

(A) Transformer
(B) Ac generator
(C) Alternator
(D) Dynamo

Q16. Power dissipated in resistive circuit for A.C.

(A) Vrms Irms
(B) VmaxImax/2
(C) Both A & B
(D) None of these

Q17. Peak value of AC is [illegible in source scan] the RMS current is.

(A) √2 A
(B) 1A
(C) 5A
(D) None of these

Q18. Which one is ferromagnetic in nature?

(A) Soft iron
(B) Nickle
(C) Copper
(D) None of these